



Module 7: Antennas & Transmission Lines

Gain · Patterns · Matching · Transmission Lines · Smith Chart · Receiving Antennas
2024-2028 Amateur Extra Question Pool — Final Module



The Final Module

- 8 exam questions — all of E9 — 16% of your Extra exam
- Tied with Module 5 as the heaviest module
- This module has the most math — but all of it builds on Module 3
- Hour 1:
 - → E9A: Antenna parameters — gain, beamwidth, efficiency, ERP, EIRP
 - → E9B: Radiation patterns and modeling software
 - → E9C: Practical wire antennas — dipole, folded dipole, arrays, rhombic, Beverage
 - → E9D: Yagi antennas and loading
- Hour 2:
 - → E9E: Impedance matching — quarter-wave section, stubs, matching networks
 - → E9F: Transmission lines — coax vs. ladder line, velocity factor, line behavior
 - → E9G: The Smith chart
 - → E9H: Receiving antennas — loop, Beverage, active



Part 1: Antenna Parameters

Subelement E9A — 1 exam question



Antenna Gain and Related Quantities

- Radiation resistance: equivalent resistance that would dissipate same power as antenna radiates
- → Half-wave dipole in free space: ~73 ohms
- → Quarter-wave vertical over perfect ground: ~35 ohms
- Antenna gain references:
 - → dBi: gain over a theoretical isotropic radiator (perfect omnidirectional)
 - → dBd: gain over a half-wave dipole
 - → Conversion: $\text{dBi} = \text{dBd} + 2.15$
- ERP (Effective Radiated Power): $\text{TX power} \times \text{antenna gain over a dipole}$
- EIRP (Effective Isotropic Radiated Power): $\text{ERP} + 2.15 \text{ dB (over isotropic)}$



Part 2: Patterns and Practical Antennas

Subelements E9B-D — 1 exam question



Radiation Patterns and Modeling

- Azimuth pattern: antenna radiation in the horizontal plane (viewed from above)
- Elevation pattern: antenna radiation in the vertical plane (side view)
- Main lobe: direction of maximum radiation
- Side lobes: unwanted radiation in other directions
- Nulls: directions of minimum radiation — useful for null-steering
- Front-to-back ratio: gain of main lobe vs. gain directly behind

- Antenna modeling software:
 - → NEC (Numerical Electromagnetics Code): industry-standard engine
 - → EZNEC and 4NEC2: graphical NEC interfaces for amateurs
 - → Models free-space patterns; real-world affected by ground and nearby structures



Practical Wire Antennas (E9C)

- Half-wave dipole: ~ 73 -ohm feed impedance in free space; classic reference antenna
- Folded dipole: ~ 300 -ohm impedance ($4\times$ regular dipole)
 - $\rightarrow$ More bandwidth; common as Yagi driven element fed through balun
- Rhombic: diamond-shaped terminated long-wire; very high gain; wide bandwidth
 - $\rightarrow$ Needs large lot; used for point-to-point HF links
- Beverage: long wire ($1+$ wavelength) close to ground, terminated
 - $\rightarrow$ RECEIVING ONLY; traveling-wave; directional; excellent low-noise on 160/80m
- Ground effects on antennas:
 - $\rightarrow$ Low dipole: high angle NVIS for regional communication
 - $\rightarrow$ High dipole ($>0.5\lambda$): low-angle radiation; better for DX
- Chordal hop (covered in Module 2) benefits from lower elevation angle = longer hops



Yagi Antennas and Loading (E9D)

- Yagi elements:
 - → Driven element: connected to feed line (approximately half-wave)
 - → Director: shorter than driven; placed IN FRONT → adds gain, narrows beam
 - → Reflector: longer than driven; placed BEHIND → increases front-to-back ratio
 - More elements = more gain + narrower beamwidth
 - Feed point impedance reduced by nearby directors and reflectors
 - Common matching methods: gamma match, T-match, hairpin match
- Parabolic dish: very high gain at microwave; feed at focal point
- Electrically short antennas and loading:
 - → Loading coil adds inductance to bring short antenna to resonance
 - → High-Q coil: narrow bandwidth, high efficiency



Part 3: Impedance Matching

Subelement E9E — 1 exam question



Quarter-Wave Matching and Stubs

- Quarter-wave matching section:
- $Z_{\text{match}} = \sqrt{Z_1 \times Z_2}$
- → A quarter-wave section with this impedance transforms Z_1 to Z_2
- → Example: match 200Ω antenna to 50Ω coax:
- $Z = \sqrt{200 \times 50} = \sqrt{10,000} = 100\Omega$
- → Use 100Ω coax (two 50Ω lines in series, or 75Ω for approximation)

- Stub matching:
- → Open or shorted transmission line sections added in parallel at feed point
- → Used to cancel reactive component
- → Shorted quarter-wave stub: appears as open circuit (high impedance) at input
- → Open quarter-wave stub: appears as short circuit at input



Part 4: Transmission Lines

Subelement E9F — 1 exam question



Transmission Line Properties

- Velocity factor (VF): ratio of signal speed in line to speed of light
 - → Always less than 1 — signal is slower in the dielectric
 - → Typical values: coax 0.66-0.78; ladder line 0.91-0.95
- Electrical length = physical length $\times$ velocity factor
 - → Free-space quarter wave at 7 MHz = 10.7m
 - → In RG-8 coax (VF=0.66): physical length = $10.7 \times 0.66 = 7.1\text{m}$
- Quarter-wave line behavior (the inversion rule):
 - → Open circuit at far end → appears as SHORT CIRCUIT at input
 - → Short circuit at far end → appears as OPEN CIRCUIT at input
- Open-wire (ladder line): lower loss than coax but balanced; needs balun



Part 5: The Smith Chart

Subelement E9G — 1 exam question



Smith Chart Layout and Reading

- The Smith chart is a graphical tool for transmission line and impedance matching problems
- Key points:
 - → CENTER: Z_0 = perfect match (SWR 1:1) — normalized value $1.0 + j0$
 - → RIGHT SIDE: infinite impedance (open circuit)
 - → LEFT SIDE: zero impedance (short circuit)
 - → UPPER HALF: inductive reactance (positive imaginary)
 - → LOWER HALF: capacitive reactance (negative imaginary)
- Lines on the chart:
 - → Constant resistance circles: pass through right side; center is their axis
 - → Constant reactance arcs: curved lines from right side
 - → SWR circles: concentric circles centered on Z_0



Using the Smith Chart

- Moving along a transmission line:
 - → Moving TOWARD GENERATOR: rotate CLOCKWISE
 - → Moving TOWARD LOAD: rotate COUNTERCLOCKWISE
 - → One complete revolution = one half wavelength along the line
- SWR circles: all points on a circle have the same SWR
 - → Center = SWR 1:1; larger circles = higher SWR
- Converting impedance to admittance:
 - → Move 180° around the chart (opposite side of same SWR circle)
- Reading a point: find R and X from the constant circles/arcs,
 - then denormalize: $Z = (R + jX) \times Z_0$



Part 6: Receiving Antennas

Subelement E9H — 1 exam question



Receiving Antennas

- Small loop antenna:
 - → Figure-8 pattern with deep nulls at right angles to the loop plane
 - → Used for Radio Direction Finding (RDF) — null points toward source
 - → Excellent for noise reduction by steering nulls at interference
- Beverage antenna:
 - → Traveling-wave antenna — 1+ wavelengths long; close to ground; terminated
 - → Pattern: cardioid (unidirectional) toward the terminated end
 - → Very low noise floor — excellent for 160m and 80m DX receiving
 - → Typical length: 500-1000 feet; terminated in 600-800 ohms
- Active receiving antenna:
 - → Small antenna element + low-noise preamplifier



Key Numbers — Module 7

dBi = dBd +	2.15
Dipole feed impedance	~73 ohms in free space
Folded dipole impedance	~300 ohms (4× dipole)
Quarter-wave match formula	$Z = \sqrt{Z1 \times Z2}$
Match 200Ω to 50Ω	$\sqrt{200 \times 50} = 100\Omega$ section
Open end of 1/4-wave line	Short circuit at input
Shorted end of 1/4-wave line	Open circuit at input
Velocity factor	Always <1; signal slower in dielectric
Electrical length	Physical length × velocity factor
Ladder line VF typical	0.91-0.95
Coax VF typical	0.66-0.78
Smith chart center	Perfect match; SWR 1:1
Smith chart right side	Open circuit
Smith chart left side	Short circuit
Smith chart upper half	Inductive reactance



Checkpoint Quiz 1 / 10

What is the impedance of a quarter-wave matching section to match 25Ω to 100Ω ?

- A. 62.5Ω
- B. 75Ω
- C. 50Ω
- D. 125Ω

Answer: C — $Z = \sqrt{25 \times 100} = \sqrt{2500} = 50\Omega$



Checkpoint Quiz 2 / 10

How does dBi relate to dBd?

- A. $\text{dBd} = \text{dBi} + 2.15$
- B. $\text{dBi} = \text{dBd} + 2.15$
- C. $\text{dBi} = \text{dBd} - 2.15$
- D. They are equal

Answer: B — $\text{dBi} = \text{dBd} + 2.15$



Checkpoint Quiz 3 / 10

What is the input impedance of a quarter-wave line with an open circuit at the far end?

- A. Open circuit
- B. Z_0
- C. Short circuit
- D. $2 \times Z_0$

Answer: C — Quarter-wave open end appears as a SHORT CIRCUIT at the input



Checkpoint Quiz 4 / 10

A Beverage antenna is best suited for:

- A. High-gain transmitting on 10m
- B. VHF direction finding
- C. Low-noise receiving on 160m and 80m
- D. Wide bandwidth transmitting

Answer: C — Beverage: receive-only, traveling-wave, low noise floor; ideal for 160/80m DX receiving



Checkpoint Quiz 5 / 10

What does the center of a Smith chart represent?

- A. Open circuit
- B. Short circuit
- C. Purely reactive impedance
- D. Perfect match (SWR 1:1)

Answer: D — Center = Z_0 normalized to $1+j0$; perfect match; SWR 1:1



Checkpoint Quiz 6 / 10

What is the approximate feed impedance of a folded dipole?

- A. 50Ω
- B. 73Ω
- C. 300Ω
- D. 600Ω

Answer: C — Folded dipole: $\sim 300\Omega$ ($4\times$ the regular dipole's $\sim 73\Omega$)



Checkpoint Quiz 7 / 10

A Yagi director is placed in front of the driven element and is:

- A. Longer; reflects energy
- B. Shorter; increases forward gain
- C. The same length; provides matching
- D. Variable length

Answer: B — Director: shorter than driven element; placed toward desired direction; increases forward gain



Checkpoint Quiz 8 / 10

What determines antenna efficiency?

- A. SWR on the feed line
- B. Ratio of radiation resistance to total resistance
- C. Resonant frequency
- D. Number of radials

Answer: B — $\text{Efficiency} = R_{\text{rad}} / (R_{\text{rad}} + R_{\text{loss}})$



Checkpoint Quiz 9 / 10

What pattern does a small loop receiving antenna produce?

- A. Omnidirectional in loop plane
- B. Figure-8 with nulls perpendicular to loop
- C. Cardioid
- D. Unidirectional toward loop opening

Answer: B — Small loop: figure-8 pattern; deep nulls perpendicular to the plane of the loop



Checkpoint Quiz 10 / 10

Moving clockwise on the Smith chart corresponds to movement in what direction on the transmission line?

- A. Toward the load
- B. Toward the generator
- C. Increasing frequency
- D. Decreasing SWR

Answer: B — Clockwise = toward generator; counterclockwise = toward load



VE Exam — You're Ready

- 50 questions, 37 to pass (74%)
- Bring: government-issued photo ID and your FRN number
- After passing: \$35 FCC fee via CORES online
- License upgrades when the FCC ULS database reflects the new grant
- Your call sign stays the same — you can apply for a 1×2 or 2×1 vanity later